

Date _____

(Q1) Discuss the geometrical interpretation of Rolle's and Lagrange's Mean value Theorem.

Ans 1 Rolle's Theorem

Statement:

● If a function $f(x)$ is

- (i) Continuous in the closed intervals $[a, b]$
- (ii) Differentiable in the open interval (a, b)
- (iii) $f(a) = f(b)$

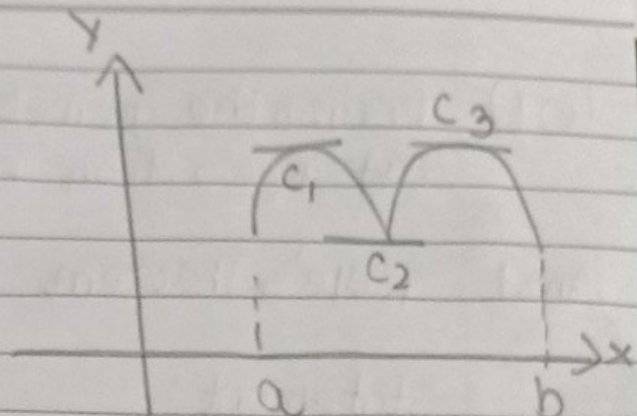
then there exists atleast one value ' c ' of ' x ' in (a, b) such that $f'(c) = 0$

Geometrical interpretation.

● Since $f(a) = f(b)$, the points $A(a, f(a))$ and $B(b, f(b))$ lie at the same height. Therefore, the line joining A and B is parallel to the x -axis.

Rolle's theorem states that there is atleast one point C between A and B where the tangent to the curve is also parallel to the x -axis.

Hence, $f'(c) = 0$



Geometrically Rolle's theorem states that the tangents c_1, c_2, c_3 are parallel to x -axis.

Lagrange's Mean Value Theorem (LMVT)

If a function $f(x)$ is

- 1) Continuous in the closed interval $[a, b]$
 - 2) Differentiable in the open interval (a, b)
- then there exist atleast one value ' c ' of ' x ' in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Geometrical Interpretation

Consider two points

$A(a, f(a)), B(b, f(b))$

on curve $y = f(x)$.

Thus, $c = 60$

The slope of the secant line AB is

$$\text{slope of AB} = \frac{f(b) - f(a)}{b - a}$$

According to LMVT, there exists atleast one point $C(c, f(c))$ between A and B, where the tangent to the curve is parallel to the secant line AB

Therefore

slope of tangent at C = slope of AB

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

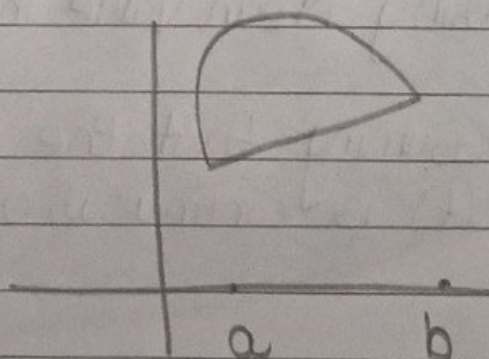
LMVT is generalisation of Rolle's Theorem

In the special case

$$f(a) = f(b)$$

then LMVT reduces to Rolle's Theorem

Geometrically LMVT states that the tangent to curve at C is parallel to the chord AB.



- (2) Let $s(t)$ be the total miles that you have driven on your trip down a toll road freeway after time t hours. Assume also that $s(1/3) = 10$ miles and $s(1.2) = 82$ miles. At a checkpoint along the toll road, you admit these facts to a highway patrol officer, who is familiar with the Mean Value Theorem. The officer makes a few quick calculations, smiles, then politely prepares to write you a traffic ticket. Explain why.

Ans 2 Let $s(t)$ den.

Let $s(t)$ denote the total distance in miles travelled by the driver after t hours.

Given:

$$s\left(\frac{1}{3}\right) = 10 \text{ miles}$$

and $s(1.2) = 82 \text{ miles}$

The speed limit on the road is 55 miles per hour. We have to determine whether the police officer is justified in issuing a traffic ticket.

Using Mean Value Theorem

Assuming that the motion of the car is continuous, $s(t)$ is continuous on $\left[\frac{1}{3}, 1.2\right]$

and differentiable on

$$\left(\frac{1}{3}, 1.2\right)$$

Therefore, by Mean Value Theorem, there exists at least one point c in this interval such that

$$s'(c) = \frac{s(1.2) - s(1/3)}{1.2 - 1/3}$$

Here $s'(c)$ represents the instantaneous speed of the car at time c .

Substituting the given values,

$$s'(c) = \frac{82 - 10}{1.2 - 1/3}$$

Since

$$1.2 = \frac{6}{5}$$

We get

$$s'(c) = \frac{72}{\frac{6}{5} - \frac{1}{3}}$$

$$s'(c) = \frac{72}{18-5}$$

$$= \frac{72}{13}$$

$$= \frac{72}{13}$$

$$= \frac{72 \times 15}{13}$$

$$= \frac{1080}{13}$$

Therefore $s'(c) = 83.08 \text{ mph}$

The permitted speed is 55 mph
but MVT guarantees that at some instant the speed
was 83.08 mph.

$$\text{so } 83.08 > 55$$

Thus, at some point between $t = \frac{1}{3}$ hour and

$t = 1.2$ hours, the driver must have been travelling
at approximately 83.08 mph, which is well
above the speed limit.

Conclusion.

Hence, according to the Mean Value Theorem, the
police officer has mathematical evidence that
the driver exceeded the speed limit during this
interval. Therefore, the officer is justified in
issuing a traffic ticket.

Q3)

A delivery of 480 cubic meters of gravel is to be made to a landfill. The trucker plans to purchase an open-top box in which to transport the gravel in numerous trips. The total cost to the trucker is the cost of the box plus Rs 80 per trip. The box must have height 2 meters, but the trucker can choose the length and width. The cost of the box is Rs 100 per square meter for the ends, Rs 50 per square meter for the sides and Rs 200 per square meter for the bottom. Notice the tradeoff. A smaller box is cheaper to buy but requires more trips. What size box should the trucker buy to minimize the total cost?

Ans 3

A delivery of 480 cubic meters of gravel is to be transported using an open-top rectangular box. The box has a fixed height of

$$h = 2 \text{ m}$$

Let

 $l = \text{length of box}$ $w = \text{width of box}$

We have to find the dimensions of the box that minimize the total cost.

Volume Constraint

Volume of a rectangular box is
 $V = lwh$

Given $V = 480 \text{ m}^3$ and $h = 2 \text{ m}$,
 $lw(2) = 480$

Therefore,

$lw = 240$
or

$l = 240$
w

The box is open at the top, so material is required only for the bottom, two sides and two ends.

Cost of bottom:

Area of bottom is lw

At ₹ 200 per square meter,

$$C_1 = 200lw$$

Cost of two sides:

Each side has area $lh = 2l$

Therefore, area of two sides is

$$2(2l) = 4l$$

At ₹ 50 per square meter,

$$C_2 = 50(4l) = 200l$$

Cost of two ends :

Each end has area

$$wh = 2w$$

Hence, area of two ends is

$$2(2w) = 4w$$

At ₹100 per square meter,

$$C_3 = 100(4w) = 400w$$

Also, there is a fixed cost of
₹ 80

Hence, total cost is

$$C = 80 + 200l + 200l + 400w$$

Since $lw = 240$,

$$C = 80 + 200(240) + 200l + 400w$$

$$C = 48080 + 200l + 400w$$

using $l = \frac{240}{w}$

We get

$$C(w) = 48080 + \frac{48000}{w} + 400w$$

Differentiate $C(w)$ with respect to w :

$$C'(w) = -\frac{48000}{w^2} + 400$$

For minimum cost,

$$C'(w) = 0$$

Therefore,

$$\frac{-48000}{w^2} + 400 = 0$$

$$w^2$$

$$400 = \frac{48000}{w^2}$$

$$w^2$$

$$w^2 = 120$$

Thus,

$$w = \sqrt{120} = 2\sqrt{30}$$

$$w = 10.95 \text{ m}$$

Now

$$l = \frac{240}{w}$$

$$l = \frac{240}{2\sqrt{30}} = 4\sqrt{30}$$

$$\text{Therefore, } l = 21.91 \text{ m}$$

Also

$$C''(w) = \frac{96000}{w^3} > 0$$

for $w > 0$. Hence, the obtained dimensions give a minimum cost.

The box should have dimensions

$$\text{Length} = 4\sqrt{30} = 21.91 \text{ m}$$

$$\text{width} = 2\sqrt{30} = 10.95 \text{ m}$$

$$\text{Height} = 2 \text{ m}$$

Thus, a box of approximately $21.91 \text{ m} \times 10.95 \text{ m} \times 2$

(Q4) Length of cantilever beam: $L=1\text{ m}$
distributed load:

$$w(x) = cx^3(1-x^2)^2 \text{ N/m}$$

Total load:

$$\int_0^1 w(x) dx = 1 \text{ N}$$

(i) Using the total load condition:

$$\int_0^1 cx^3(1-x^2)^2 dx = 1$$

$$(1-x^2)^2 = 1 - 2x^2 + x^4$$

Therefore,

$$cx^3(1-2x^2+x^4) = c(x^3-2x^5+x^7)$$

$$c \int_0^1 (x^3 - 2x^5 + x^7) dx = 1$$

$$c \left[\frac{x^4}{4} - \frac{2x^6}{6} + \frac{x^8}{8} \right]_0^1 = 1$$

Putting $x=1$:

$$c \left(\frac{1}{4} - \frac{1}{3} + \frac{1}{8} \right) = 1$$

Taking LCM

$$c \left(\frac{6-8+3}{24} \right) = 1$$

$$C \left(\frac{1}{24} \right) = 1$$

$$C = 24$$

So load distribution is

$$w(x) = 24x^3(1-x^2)^2$$

(2) find centroid $\bar{x}$

$$\bar{x} = \frac{\int_0^1 x w(x) dx}{\int_0^1 w(x) dx}$$

$$\int_0^1 w(x) dx = 1$$

Therefore,

$$\bar{x} = \int_0^1 x w(x) dx$$

substitute $w(x)$:

$$\bar{x} = \int_0^1 x [24x^3(1-x^2)^2] dx$$

$$= 24 \int_0^1 x^4 (1-2x^2+x^4) dx$$

$$= 24 \int_0^1 (x^4 - 2x^6 + x^8) dx$$

Integrating:

$$\bar{x} = 24 \left[\frac{x^5}{5} - \frac{2x^7}{7} + \frac{x^9}{9} \right]_0^1$$

$$= 24 \left(\frac{1}{5} - \frac{2}{7} + \frac{1}{9} \right)$$

$$= 24 \left(\frac{63 - 90 + 35}{315} \right)$$

$$= 24 \left(\frac{8}{315} \right) = \frac{192}{315} = \frac{64}{105}$$

$$\bar{x} = \frac{64}{105} \text{ m}$$

$$\bar{x} \approx 0.61 \text{ m}$$

$$C = 24$$

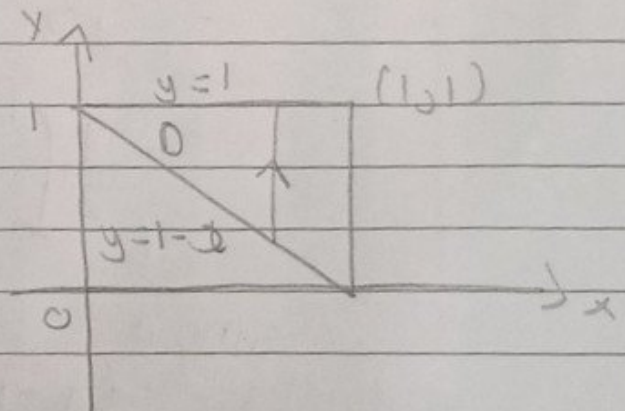
$$\bar{x} = \frac{64}{105} \text{ m} \approx 0.61 \text{ m}$$

so the equivalent point load of 1 N acts approximately 0.61 m from the fixed end.

(Q5) Total charge

Given charge density:

$$\sigma(x, y) = xy \text{ C/m}^2$$



From the diagram, triangular region is bounded by

$$y=1; \quad y=1-x, \quad x=1$$

So the limits are:

$$0 \leq x \leq 1;$$

$$1-x \leq y \leq 1$$

Formula for total charge

$$Q = \iint_D \sigma(x, y) dA$$

substitute $\sigma(x, y) = xy$:

$$Q = \int_0^1 \int_{1-x}^1 xy \, dy \, dx$$

Integrate with respect to y

$$Q = \int_0^1 x \left[\frac{y^2}{2} \right]_{1-x}^1 dx$$

$$= \frac{1}{2} \int_0^1 x [1 - (1-x)^2] dx$$

since

$$(1-x)^2 = 1 - 2x + x^2$$

therefore,

$$Q = \frac{1}{2} \int_0^1 x(2x - x^2) dx$$

$$= \int_0^1 \left(x^2 - \frac{x^3}{2} \right) dx$$

Integrate

$$Q = \left[\frac{x^3}{3} - \frac{x^4}{8} \right]_0^1$$

$$= \frac{1}{3} - \frac{1}{8}$$

$$= \frac{8-3}{24}$$

$$= \frac{5}{24} \text{ C}$$

$$\text{Total charge} = \frac{5}{24} \text{ C} \approx 0.2083 \text{ C}$$